

QUESTION 1 [16 marks]

- a. **(10 marks)** For the circuit shown in Figure 1, calculate the equivalent resistance R_{eq} as seen from terminal $a-b$.

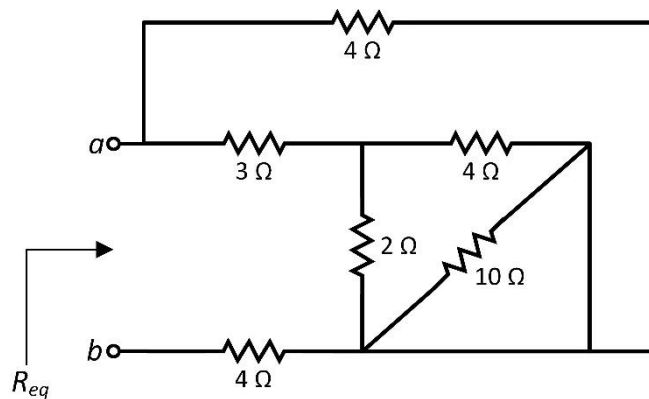


Figure 1

- b. **(6 marks)** The two circuits in Figure 2 show two different configurations for connecting multiple light-emitting diodes (LEDs) to a battery with an internal resistance R . The forward biasing voltage of the LEDs is $V_f = 1.5V$, the battery voltage is $V_s = 5V$ and the battery internal resistance value is $R = 0.1\Omega$.
- (4 marks)** Calculate the current flowing through the battery in both configurations, i.e. i_{s1} and i_{s2} .
 - (2 marks)** Which of these configurations will discharge the battery the quickest? Explain why this is the case.

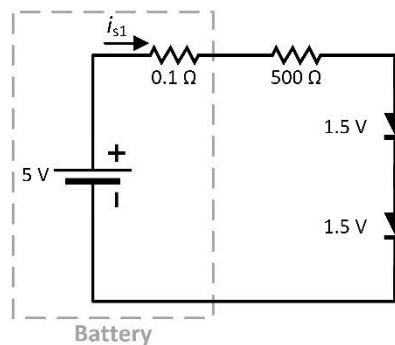


Figure 2 (a)

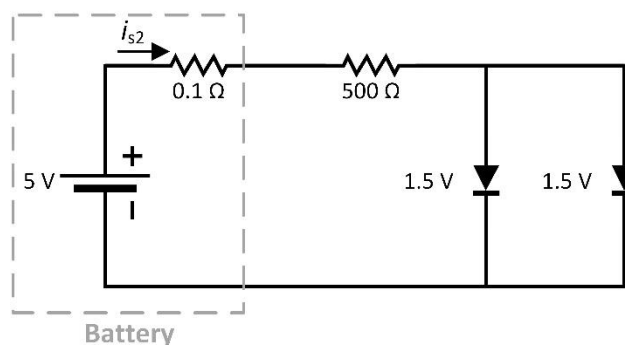


Figure 2 (b)

QUESTION 2 [20 marks]

The circuit in Figure 3 represents a solar-powered caravan. The caravan has a solar panel on the roof to power a fridge and charge a battery during the day.

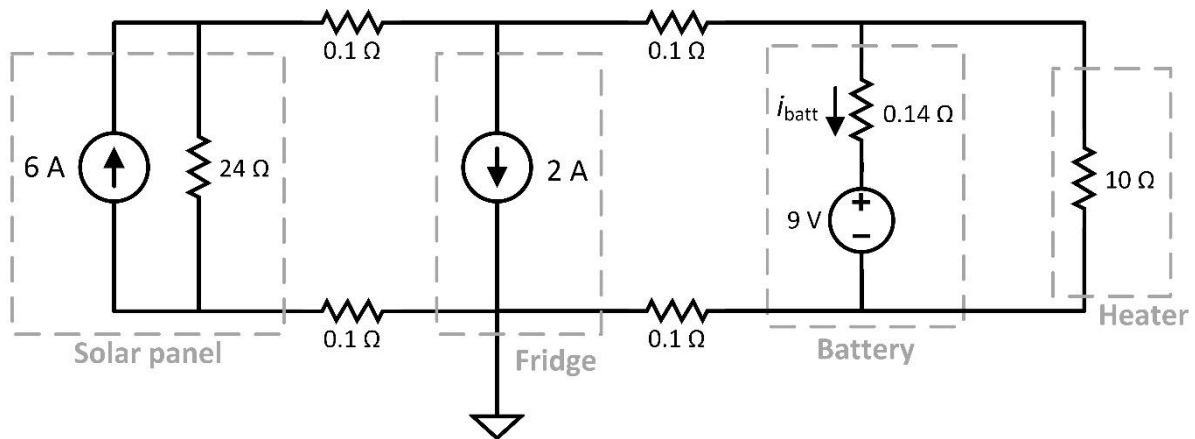


Figure 3

- (15 marks)** Perform superposition to find the current flowing through the battery (i_{batt}) during the day (when the solar panel is generating power).
- (5 marks)** Find the power in the battery.

QUESTION 3 [24 marks]

- a. **(12 marks)** For the circuit shown in Figure 4, apply nodal analysis and write down the nodal equations using the labels provided. Note: Simplify the equations in the form $aV_1 + bV_2 + cV_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but DO NOT solve them.

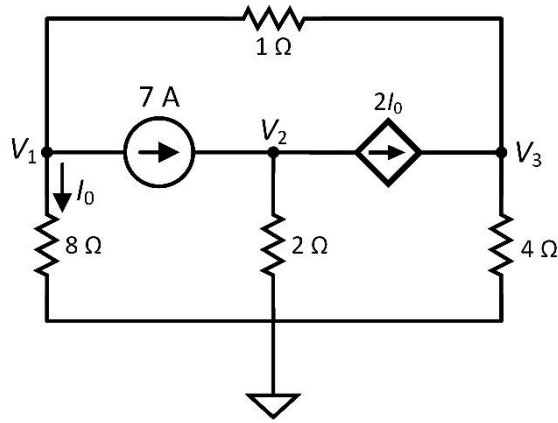


Figure 4

- b. **(12 marks)** For the circuit shown in Figure 5, apply mesh analysis and write down the mesh equations using the labels provided. Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but DO NOT solve them.

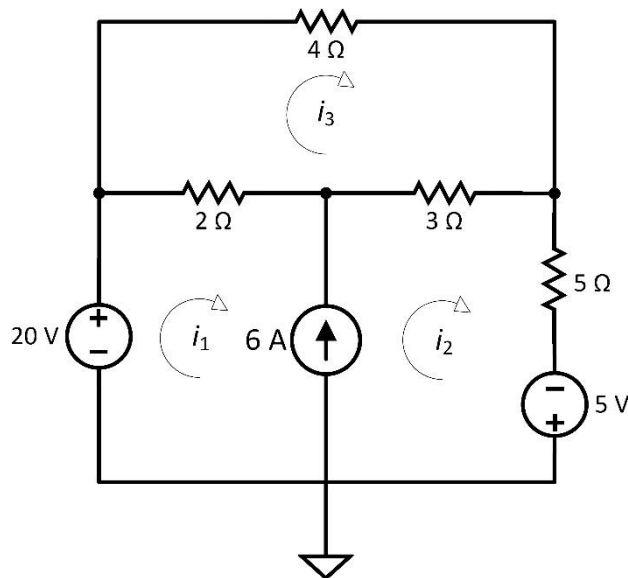


Figure 5

QUESTION 4 [20 marks]

The circuit shown in Figure 6 is being used to power a purely resistive load which can be modelled as a single resistor R_L connected to terminals a - b .

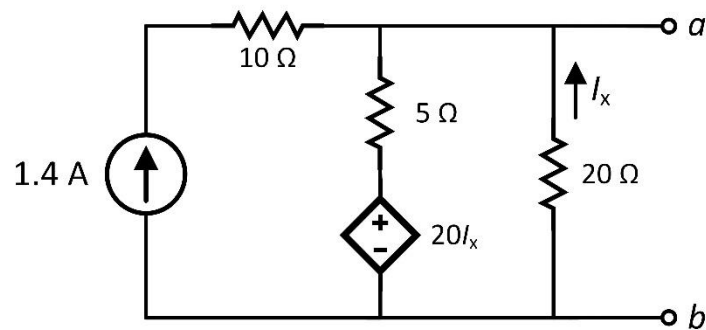


Figure 6

- (9 marks)** Calculate the Thevenin voltage at terminals a - b .
- (9 marks)** Calculate the load resistance that will ensure the maximum transfer of power from the circuit to the heater.
- (2 marks)** Find the maximum power that can be delivered.

QUESTION 5 [20 marks]

- a. **(8 marks)** Find the voltage across the capacitors in the circuit of Figure 7 under DC steady state conditions.

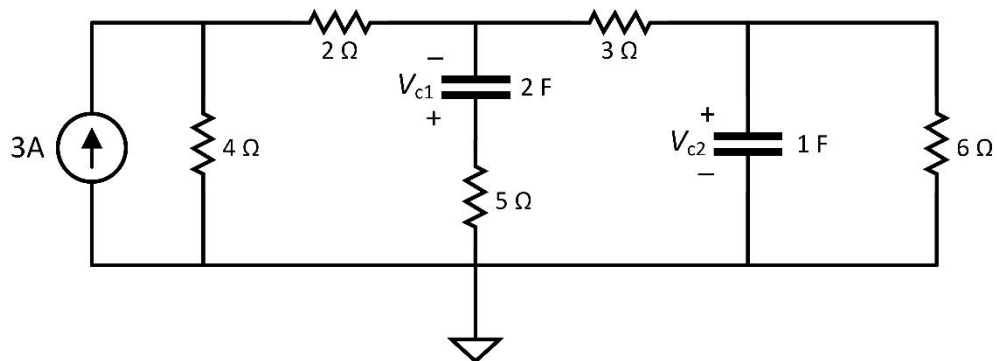


Figure 7

- b. **(12 marks)** The circuit in Figure 8 is used to control a flashing light. The lamp in this circuit starts to conduct whenever the lamp voltage reaches 10 V. During the time the lamp conducts, it can be modeled as a $10\text{ k}\Omega$ resistor. Once the lamp conducts, it will continue to conduct until the lamp voltage drops to 2 V. When the lamp is not conducting, it behaves as an open circuit.

Assume that the circuit has been in operation for a long time. Let $t = 0\text{ s}$ at the instant when the lamp starts conducting. Thus, at $t = 0\text{ s}$ the lamp is modeled as a $10\text{ k}\Omega$ resistor, and the voltage drop across the lamp is 10 V.

Find the time instant when the lamp will stop conducting (i.e. time instant when lamp voltage drops to 2 V and stops behaving as a $10\text{ k}\Omega$ resistor).

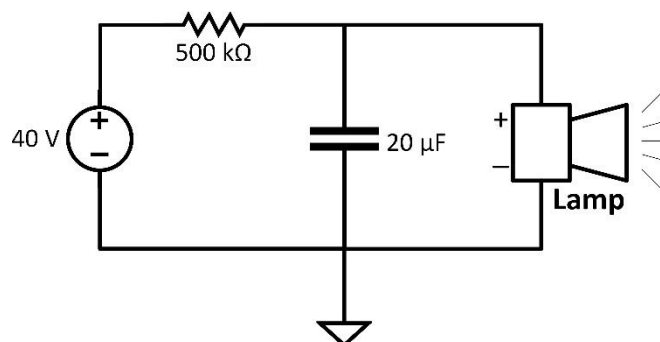


Figure 8